

Improving Downstream Task Performance in Multilingual LLMs by Intervening Language Specific Neurons

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Introduction

Problem Statement

• Objective:

- Investigate the representation of language-specific information in large language models (LLMs).
- Develop methods to identify and utilize language-specific neurons for multilingual tasks.
- Enhance performance for low-resource languages by employing techniques such as activation probability entropy and LoRA-based fine-tuning.

• Importance:

- Ensure fair and inclusive NLP systems by bridging the performance gap between high-resource and low-resource languages.
- Advance understanding of the internal mechanisms of multilingual LLMs.
- Contribute to the development of efficient methods for improving multilingual performance.

Presentation Outline

- Review of existing methods for identifying and utilizing language-specific neurons.
- Experimental design: fine-tuning configurations and intervention strategies.
- Results and analysis for multilingual tasks (e.g., XNLI task).
- Conclusion and future work directions.

Methodology: Definition of Neuron

What is a Neuron?

- A **neuron** corresponds to a specific dimension of the activation vector $\mathbf{h}^i$, represented as:

$$\mathbf{h}^i = [\text{act_fn}(\hat{\mathbf{H}}^i \mathbf{W}_1^i) \otimes (\hat{\mathbf{H}}^i \mathbf{W}_3^i)] \cdot \mathbf{W}_2^i, \quad (1)$$

- The average activation (against Wikimedia sentence) of a neuron (i, j) across token positions is computed as:

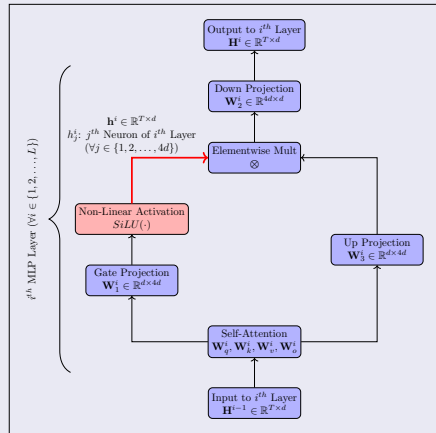
$$\bar{h}_{i,j} = \frac{1}{T} \sum_{t=1}^T h_{i,j}^t. \quad (2)$$

- Language-specific activations are computed as:

$$h_{i,j}^l = \frac{1}{|S_l|} \sum_{s \in S_l} \bar{h}_{i,j}^{(s)}, \quad (3)$$

where S_l is the set of all sequences for language l .

MLP Architecture of Llama3 [Grattafiori et al.]



Methodology: LAPE

LAPE Neurons [Tang et al.]

- To identify neurons with significant language-specific behavior, a threshold (for Wikipedia dataset) is defined as:

$$\tau = \text{Quantile}_{0.95} \left(p_{i,j}^l \mid \forall i, j, l \right). \quad (4)$$

- A neuron (i, j) is considered as language neuron if:

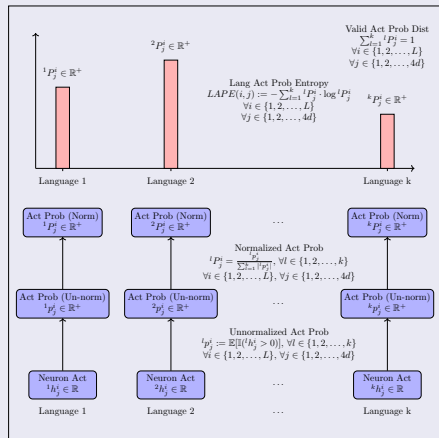
$$\exists l \in \{1, \dots, k\} \text{ such that } p_{i,j}^l > \tau \quad (5)$$

- Only 1% ($m = 0.01 \times 4d \times L$) of neurons with lowest LAPE values are selected for as language neurons.

$$LNS = \text{argsort}_m \left\{ \text{LAPE}(i, j) \mid \forall i, j, \exists l \text{ s.t. } p_{i,j}^l > \tau \right\} \quad (6)$$

$$\text{NTL}(i, j) = \left\{ l \mid p_{i,j}^l > \tau \wedge (i, j) \in LNS \right\} \quad (7)$$

LAPE Workflow



Methodology: Activation Based Methods

Neuron Relevance

- **Activation Probability 90p:** Measures the probability of a neuron's activation exceeding the 90th percentile value for its layer.
- Relevance for a neuron (i, j) in language l is computed from its activation against Wikipedia dataset:

$$P_q(h_{i,j}^l) = \inf \left\{ x \in \mathbb{R} : CDF_{h_{i,j}^l}(x) \geq q \right\} \quad (8)$$

$$r_{i,j}^l = \mathbb{E} \left[\mathbb{I} \left(h_{i,j}^l > P_{0.9}(h_{i,j}^l) \right) \right] \quad (9)$$

- Aggregates relevance values across neurons:

$$R^l = \{r_{i,j}^l \mid i \in \{1, \dots, L\}, j \in \{1, \dots, 4d\}\}. \quad (10)$$

Activation Based Neurons

- **Selection Process:** Identify top neurons with the highest relevance for each language.
- Select top 0.125% neurons per language based on their maximum relevance:

$$m = \frac{0.125 \times L \times 4d}{100} \quad (11)$$

$$LTN(l) = \text{argsort_desc}_m \left\{ r_{i,j}^l \mid \forall i, j \right\} \quad (12)$$

- This process ensures that neurons with the highest language-specific relevance are identified for each language.

Methodology: Neuron Fine-Tuning using LoRA

LoRA Fine-Tuning for a Task

- **LoRA Mechanism:** Fine-tunes a subset of parameters in MLP by low-rank decomposition [Hu et al.]:

$$[\text{masked_A}]_{ij} = 1, \forall i \in \{1, \dots, d\}, \forall j \in \{1, \dots, r\} \quad (13)$$

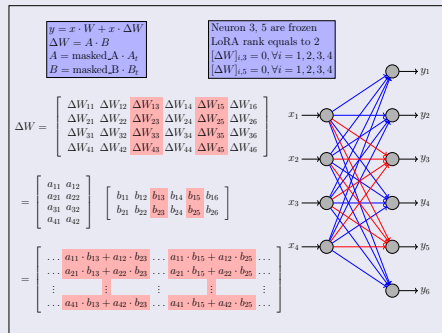
$$[\text{masked_B}]_{ij} = \begin{cases} 0, & \text{if } j \in \text{Frozen Neurons,} \\ 1, & \text{otherwise.} \end{cases} \quad (14)$$

- To freeze specific neurons, columns of B corresponding to the frozen neurons are masked with zeros. This prevents any updates to these neurons during fine-tuning.

$$\Delta W_{i,j} = 0, \quad \forall i \in 1, \dots, d, \forall j \in \text{Frozen Neurons} \quad (15)$$

- LoRA fine-tunes the MLP layer for the trainable language neurons and attention layer (q, k, v, o projections). The task-specific classifier head is fully trained.

Targeted LoRA Fine-Tuning Example



Experiments: Dataset and Baseline

Task and Dataset: XNLI

- **XNLI Task:** Multilingual natural language inference task to classify relationships between premise and hypothesis into *entailment*, *neutral*, or *contradiction* [Conneau and Others].
- **Dataset:** Sentence pairs annotated across 15 languages. Focused on English (*en*) as source language and Vietnamese (*vi*), Hindi (*hi*), and Urdu (*ur*) as target languages.
- For the experiments, a fraction $\mathcal{F}$ is used to train the model as $|\mathcal{D}_l^{\text{train}}| = 400K$.

$$|\mathcal{F}| = \text{frac} \times |\mathcal{D}_l^{\text{train}}| \quad (16)$$

where $\text{frac} = 0.25$ is the fraction of the training dataset used.

- **Objective:** Improving the zero-shot performance for low-resource languages (*vi*, *hi*, *ur*).

Baseline Experiments (No Eval Time Interventions!)

- **Zero-Shot Baseline:** Fine-tune on English (*en*) XNLI dataset. Evaluate zero-shot transfer to target languages (*vi*, *hi*, *ur*):

$$\text{Accuracy} = \frac{\sum_{i=1}^N \mathbb{I}(\hat{y}^i = y^i)}{N} \quad (17)$$

where N is the total test examples.

- **Direct Fine-Tuning Baseline:** Fine-tune on target language such as Vietnamese (*vi*) XNLI dataset to establish the upper bound for *vi* performance. Accuracy remains the evaluation metric.

Experiments: Goals and Fine-Tuning Strategy

Experiment Goals

- **Objective:** Improve zero-shot performance on target language (*vi*) without using target language training data.
- **Constraints:**
 - No target language training data (*vi*) allowed.
 - Only English (*en*) XNLI training data available.
 - Use model surgery techniques (e.g., neuron fine-tuning, activation pattern interventions).

- **Target Metric:**

$$\Delta_{\text{ZS}} = \text{Accuracy}(\mathcal{M}_{\text{experiment}}) - \text{Accuracy}(\mathcal{M}_{\text{en}}) \quad (18)$$

- Maximize Δ_{ZS} to close the performance gap between zero-shot and direct fine-tuning.

Fine-Tuning Setups

- **Setup A:** Standard fine-tuning using LoRA applied to q, k, v, o projection matrices and the classification head. Evaluated with statistical activation substitutions ($\mu, 90p, 95p$, etc.).
- **Setup B:** Fine-tuning with LoRA targeting English-specific neurons ($\mathcal{N}_{\text{en}}$) in the MLP block.
- **Setup C:** Fine-tuning with LoRA targeting Vietnamese-specific neurons ($\mathcal{N}_{\text{vi}}$) in the MLP block.
- **Setup D:** Combined fine-tuning targeting both English ($\mathcal{N}_{\text{en}}$) and Vietnamese ($\mathcal{N}_{\text{vi}}$) neurons with LoRA.

Experiments: Neuron Distribution and Intervention

Language Neuron Distribution

- The number of language-specific neurons for a language l is computed as:

$$N^l = \sum_{i=1}^L \sum_{j=1}^{4d} \mathbb{I}(p_{i,j}^l > \tau \wedge (i,j) \in LNS) \quad (19)$$

- Neuron overlap between two languages l_1 and l_2 is given by:

$$O_{l_1, l_2} = \sum_{i=1}^L \sum_{j=1}^{4d} \mathbb{I}(p_{i,j}^{l_1} > \tau \wedge p_{i,j}^{l_2} > \tau \wedge (i,j) \in LNS) \quad (20)$$

- Layer-wise neuron distribution for a language l is calculated:

$$L_l(i) = \sum_{j=1}^{4d} \mathbb{I}(p_{i,j}^l > \tau \wedge (i,j) \in LNS) \quad (21)$$

Perplexity Change

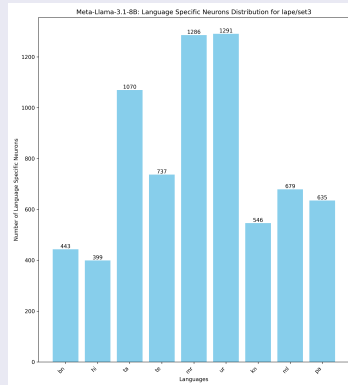
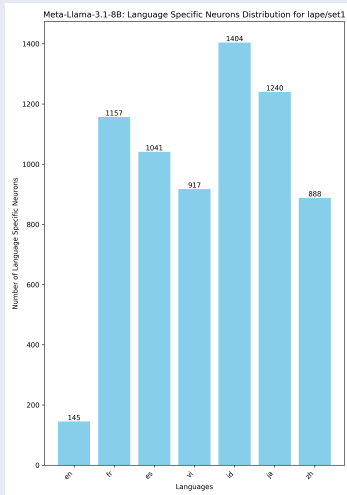
- Perplexity measures the uncertainty of a model's predictions, defined as:

$$\text{PPX}(\mathbf{x}) = \exp \left(-\frac{1}{T} \sum_{t=1}^T \log P(x_t | x_{<t}) \right) \quad (22)$$

- Perplexity change ($\text{PPXC}(i, j)$) quantifies the impact of interventions at language i on language j :

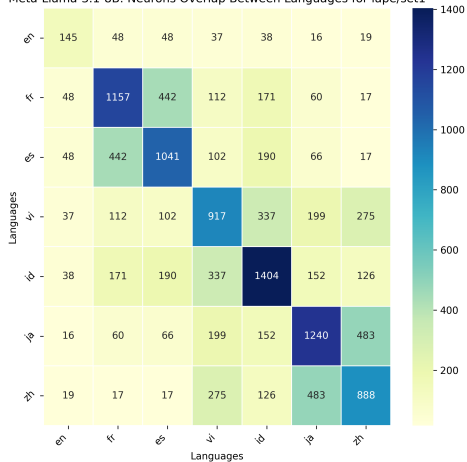
$$\begin{aligned} \text{PPXC}(i, j) = & \text{PPX}(j | \text{Intervention at } i) \\ & - \text{PPX}(j | \text{No Intervention at } i) \end{aligned} \quad (23)$$

Results: Language Neuron Distribution for LAPE

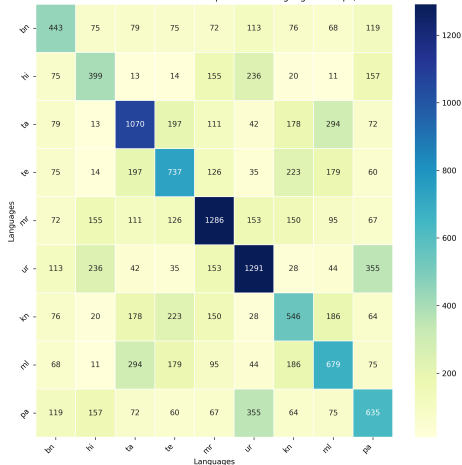


Results: Language Neuron Overlap for LAPE

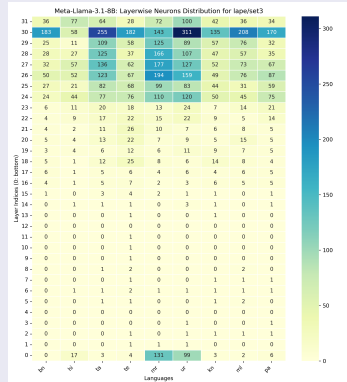
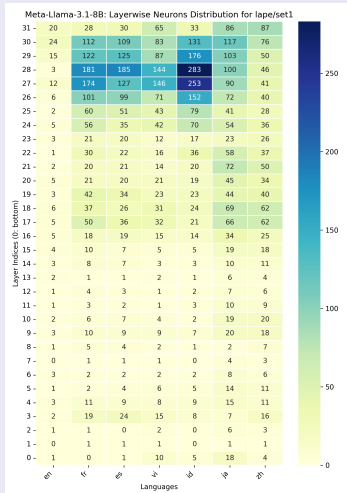
Meta-Llama-3.1-8B: Neurons Overlap Between Languages for lape/set1



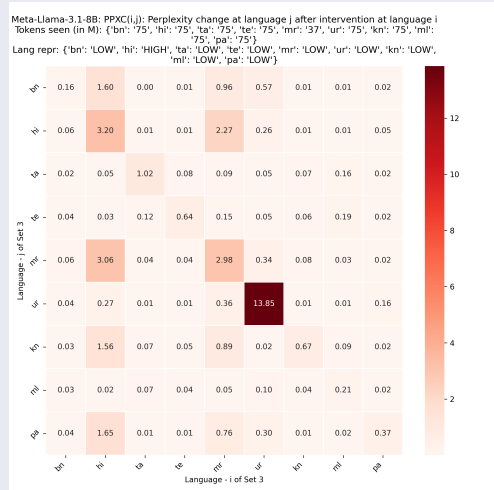
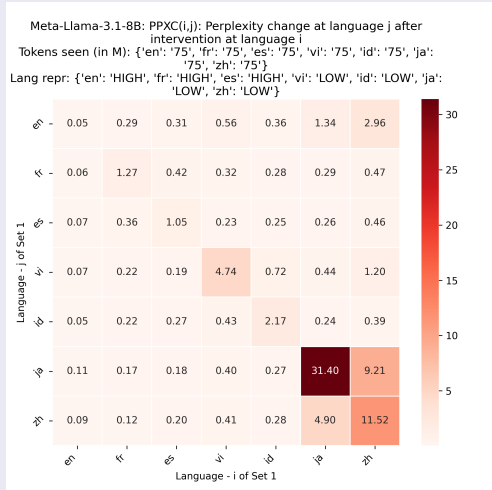
Meta-Llama-3.1-8B: Neurons Overlap Between Languages for lape/set3



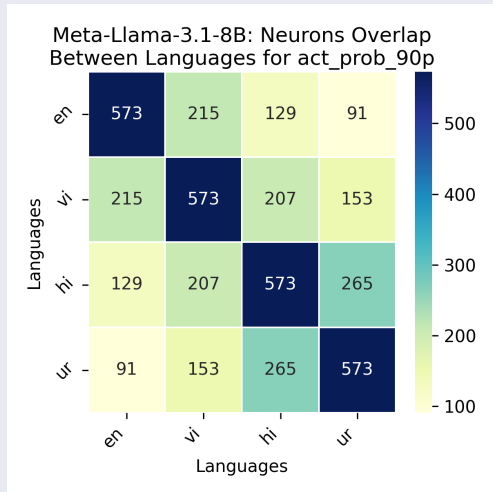
Results: Layer-wise Neuron Distribution for LAPE



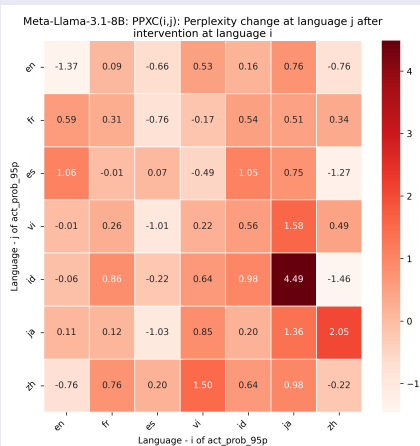
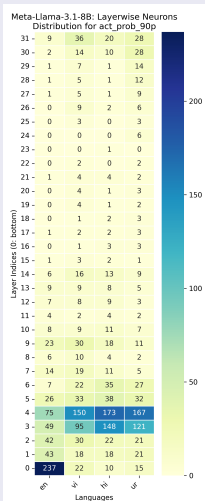
Results: Perplexity Change for LAPE



Results: Language Neuron Overlap for Activation Based Method



Results: Layer-wise Dist & Perplexity for Activation Based Method



Results: Fine-Tuning of Baseline for LAPE

XNLI Accuracy Results

Baseline Type	Train Lang	Eval Lang	No Int Accuracy
Zero Shot	en	vi	0.8053
	en	hi	0.7501
	en	ur	0.7002
Direct Fine-tuning	vi	vi	0.8511
	hi	hi	0.8043
	ur	ur	0.6697

Analysis

- The observation that direct fine-tuning for Urdu (ur) yields lower accuracy compared to zero-shot transfer because the dataset used for fine-tuning might be of low quality.
- The pretrained model's under-representation of Urdu during training might also limit the efficacy of fine-tuning.
- In contrast, zero-shot transfer utilizes cross-lingual embeddings and the model's generalized multilingual capabilities, enabling better performance.

Results: Targeted Neuron Fine-Tuning for LAPE

XNLI Accuracy Results for Intervention by Higher Quantiles

FT Neurons	Eval Lang	No Int	Int by μ	Int by P75	Int by P90	Int by P95
null	vi	0.8053	0.7955	0.7915	0.7903	0.7897
set1_en	vi	0.8017	0.7957	0.7867	0.7847	0.7847
set1_vi	vi	0.8007	0.7951	0.7881	0.7855	0.7845
set1_en+vi	vi	0.8009	0.7943	0.7865	0.7845	0.7833
null	hi	0.7501	0.7521	0.7496	0.7499	0.7506
set1_en	hi	0.7476	0.7456	0.7454	0.7448	0.7434
set6_en	hi	0.7491	0.7456	0.7464	0.7458	0.7441
set6_hi	hi	0.7492	0.7456	0.7461	0.7448	0.7431
set6_en+hi	hi	0.7488	0.7448	0.7451	0.7446	0.7431
null	ur	0.7002	0.7044	0.7028	0.6938	0.6904
set6_en	ur	0.6984	0.7041	0.7026	0.6964	0.6923
set6_ur	ur	0.6984	0.7054	0.7024	0.6952	0.6913
set6_en+ur	ur	0.6992	0.7061	0.7021	0.6954	0.6919

Results: Targeted Neuron Fine-Tuning for LAPE

XNLI Accuracy Results for Intervention by Lower Quantiles

FT Neurons	Eval Lang	No Int	Int by P5	Int by P10	Int by P25	Int by P50
null	vi	0.8053	0.7366	0.7773	0.8055	0.7971
set1_en	vi	0.8017	0.7432	0.7803	0.8027	0.7923
set1_vi	vi	0.8007	0.7426	0.7801	0.8029	0.7927
set1_en+vi	vi	0.8009	0.7426	0.7805	0.8027	0.7919
null	hi	0.7501	0.7488	0.7518	0.7496	0.7526
set1_en	hi	0.7476	0.7451	0.7468	0.7461	0.7451
set6_en	hi	0.7491	0.7464	0.7456	0.7476	0.7468
set6_hi	hi	0.7492	0.7461	0.7456	0.7464	0.7456
set6_en+hi	hi	0.7488	0.7456	0.7468	0.7462	0.7454
null	ur	0.7002	0.6861	0.6873	0.6971	0.7054
set6_en	ur	0.6984	0.6886	0.6904	0.7016	0.7031
set6_ur	ur	0.6984	0.6873	0.6913	0.7011	0.7034
set6_en+ur	ur	0.6992	0.6867	0.6893	0.7016	0.7042

Results: Targeted Neuron Fine-Tuning for Act_Prob_90p

XNLI Accuracy Results for Intervention by Higher Quantiles

FT Neurons	Eval Lang	No Int	Int by μ	Int by P75	Int by P90	Int by P95
null	vi	0.8053	0.7821	0.7891	0.7931	0.7957
en	vi	0.8027	0.7801	0.7865	0.7907	0.7923
vi	vi	0.8005	0.7799	0.7857	0.7901	0.7907
en+vi	vi	0.8005	0.7799	0.7853	0.7897	0.7909
null	hi	0.7501	0.7411	0.7268	0.7186	0.7122
en	hi	0.7492	0.7386	0.7262	0.7214	0.7148
hi	hi	0.7486	0.7382	0.7262	0.7212	0.7148
en+hi	hi	0.7488	0.7372	0.7256	0.7214	0.7144
null	ur	0.7002	0.6971	0.6978	0.6932	0.6875
en	ur	0.6994	0.7006	0.6978	0.6937	0.6893
ur	ur	0.6988	0.7001	0.6982	0.6935	0.6925
en+ur	ur	0.6978	0.6992	0.6961	0.6927	0.6915

Results: Targeted Neuron Fine-Tuning for Act_Prob_90p

XNLI Accuracy Results for Intervention by Lower Quantiles

FT Neurons	Eval Lang	Int by P5	Int by P10	Int by P25	Int by P50
null	vi	0.7653	0.7747	0.7801	0.7835
en	vi	0.7667	0.7749	0.7789	0.7827
vi	vi	0.7653	0.7743	0.7779	0.7817
en+vi	vi	0.7663	0.7745	0.7795	0.7805
null	hi	0.7501	0.7468	0.7422	0.7378
en	hi	0.7481	0.7451	0.7426	0.7378
hi	hi	0.7486	0.7456	0.7416	0.7374
en+hi	hi	0.7491	0.7462	0.7428	0.7364
null	ur	0.6948	0.6958	0.6954	0.6966
en	ur	0.6948	0.6946	0.6956	0.7001
ur	ur	0.6946	0.6952	0.6952	0.6998
en+ur	ur	0.6942	0.6964	0.6948	0.6998

Results: Targeted XNLI Task Neuron Fine-Tuning for Act_Prob_90p

XNLI Accuracy Results for Intervention by Higher Quantiles

FT Neurons	Eval Lang	No Int	Int by μ	Int by P75	Int by P90	Int by P95
null	vi	0.8053	0.7926	0.7994	0.8014	0.8028
en	vi	0.7956	0.7926	0.8004	0.8006	0.8026
vi	vi	0.7951	0.7926	0.7998	0.7998	0.8014
en+vi	vi	0.7921	0.7924	0.7982	0.7921	0.8012

XNLI Accuracy Results for Intervention by Lower Quantiles

FT Neurons	Eval Lang	Int by P5	Int by P10	Int by P25	Int by P50
null	vi	0.7706	0.7801	0.7912	0.7936
en	vi	0.7702	0.7791	0.7896	0.7926
vi	vi	0.7688	0.7774	0.7896	0.7931

Conclusions and Future Directions

Conclusions

- Statistical activation patterns (e.g., mean, P75, P90, P95) yielded limited consistent improvements across configurations.
- Language-specific neurons derived from Wikipedia data provide strong baselines for zero-shot transfer.
- Zero-shot transfer performance occasionally surpassed direct fine-tuning, particularly in languages like Urdu.
- Interventions using lower quantiles (e.g., P5, P10) degraded performance, while higher quantiles were more effective.

Future Directions

- Investigate neuron selection at the attention layer for improved fine-tuning strategies.
- Analyze cross-lingual generalization impacts, especially for low-resource languages.
- Scale methods to larger architectures like Meta-Llama-13B for insights into scalability.
- Explore interpretability to understand why zero-shot transfer sometimes outperforms direct fine-tuning.

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Thank You!